

EK PHOTO DUKAAN

Photograph the shelf. Payments keep the register.

A Paytm-for-Business add-on that turns one photo into a catalog, then turns every payment amount into a proposed basket and a smarter restock.

No typing

No new hardware

Merchant confirms every guess

No one will type 200 SKUs into an app.

At 20 seconds per item, setup alone takes more than an hour. Prices change. The register goes stale before it becomes useful.

She guesses what is left

Stock is a range in her head, not a database.

She discovers stockouts from customers

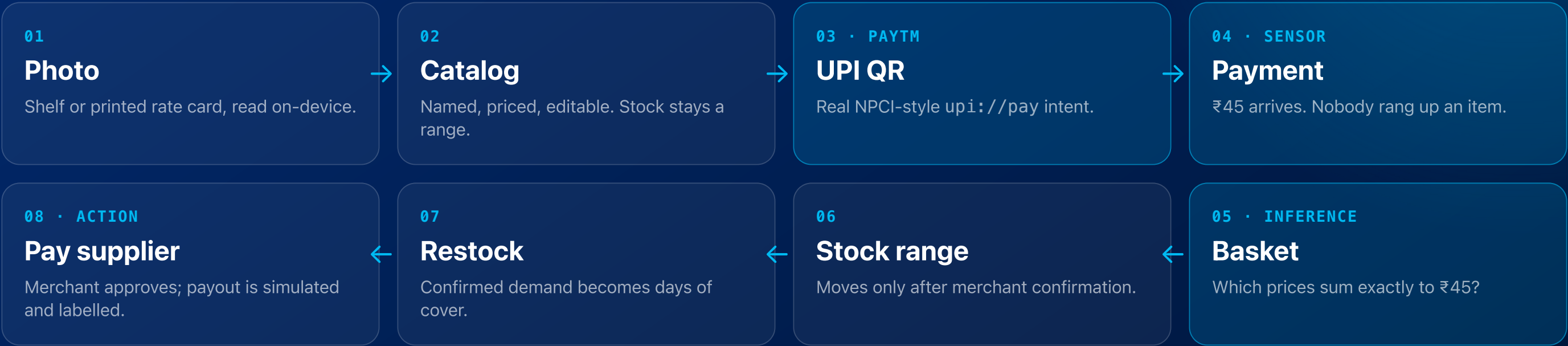
The fast mover is gone on Saturday evening.

Money arrives, but “what sold?” does not

A ₹45 payment carries an amount, not a barcode.

The cold-start problem is not intelligence. It is unpaid typing.

One photo starts it. The payments she already takes keep it running.



🔄 **Closed loop:** money in observes demand; money out acts on it.

Remove Paytm and there is no transaction stream to observe and no supplier payment to complete the loop.

LIVE OCR PROOF · MEASURED ON THE SHIPPED RATE-CARD PHOTO

Real text in.
No invented rows out.

13 94% 12/12

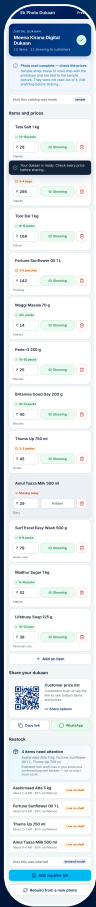
text lines read mean OCR confidence priced rows kept

The 13th line was the heading. It had no price, so the parser dropped it and showed why.

On-device OCR

Per-row confidence

Merchant edits



Real app screen · every item remains editable

A cluttered shelf photo returns zero readable lines and the app refuses. No silent fallback catalog.

The amount becomes a proposed basket.

12 items → search 14 combinations → 1 exact basket

₹45

payment amount

1×

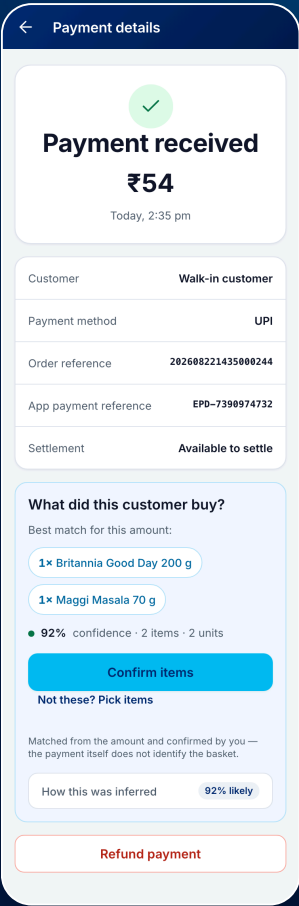
Thums Up 750 ml

92%

heuristic cap, not
accuracy

The model proposes. Meena confirms. Only then does the
stock range move.

If several baskets fit, alternates are shown and confidence drops. If none fit, the app asks for manual input.



Real app screen · 14 searched, one
exact basket

Intelligence that refuses instead of guessing.

01 · PERCEPTION

On-device OCR

Tesseract.js reads printed text in the browser. No upload, API key, or per-call fee.

02 · RESOLUTION

Fuzzy name match

Repairs likely spelling errors against a product lexicon without adding unseen products.

03 · INFERENCE

Bounded subset-sum

Searches exact-price baskets within strict item, quantity, and compute limits.

04 · PREDICTION

Weighted demand

Confirmed baskets become velocity, days of cover, and reorder suggestions.

No text?

No catalog.

No exact sum?

Ask the merchant.

Several sums?

Show choices; never decrement silently.

Rule-based, bounded, and auditable. Every displayed confidence has a stated ceiling.

Money in is the sensor. Money out is the action.

REAL · CUSTOMER SIDE

UPI QR + intent

A standards-style `upi://pay` URI opens a real UPI app with validated fields.

REAL · MERCHANT SIDE

WhatsApp draft

A real `wa.me` deep link opens a pre-filled message. The merchant presses send.

FEASIBLE · ADAPTER SEAM

Live rails can replace one layer

Payment I/O sits behind a documented service seam and webhook design.

REAL AND RUNNING

OCR, solver, forecast, catalog persistence

UPI intent construction and scannable QR

Merchant-approved basket and order workflow

SIMULATED AND LABELLED

Payment authorisation and settlement

Paytm webhook receipt

Supplier payout; no bank is contacted

Independent hackathon prototype. Not affiliated with or endorsed by Paytm. No live settlement is claimed.

IMPACT + FEASIBILITY · DESIGNED TO SHIP AS AN ADD-ON

Camera onboarding. Zero marginal AI cost.

No second app

Lives under "More for your business" in a Paytm-for-Business merchant shell.

No new hardware

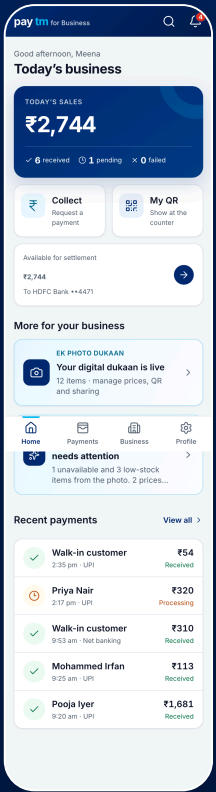
Phone camera, existing QR, existing merchant workflow.

Runs on the merchant's device

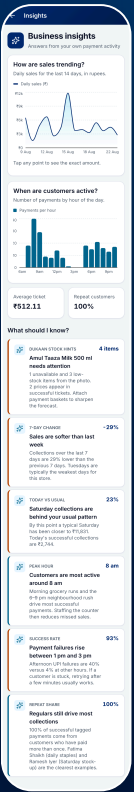
OCR and inference need no GPU and add no per-merchant model bill.

Verified path

Over 150 tests plus a 40/40 judge harness.



Add-on inside the merchant shell



Ranges and days of cover, not fake counts

Pilot metric: stockout days per fast-moving SKU. No pilot result is claimed today.

Built for the small catalog Meena can review herself.

SCOPE

10–15 items today

Not a 2,000-SKU supermarket. Distinct prices make amount inference useful.

INPUT

Printed type today

Handwriting is not the judged path. Weak reads show confidence and stay editable.

BLIND SPOT

Cash is invisible

Paytm sales are observed. Cash cannot be claimed as known.

TRY IT

merchant-app-black.vercel.app

INSPECT IT

github.com/icvasu/paytm-ek-photo-dukaan

One photo to start. After that, her own payments keep her books.

Ek Photo Dukaan is an independent hackathon prototype. Payment authorisation, settlement, webhooks, and supplier payout are simulated and labelled. No real money moves.